Emergent $\mathcal{PT}$ Symmetry and Exceptional Points in a Driven Dirac Impurity

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Periodic driving can produce passive non-Hermitian impurity dynamics without microscopic gain. We derive this mechanism for an inversion-asymmetric spin–orbit-coupled (SOC) Dirac impurity. Off-shell angular harmonics generate a real spin-odd shift, which the retarded matrix bath converts into relative spin-dependent decay. After common damping is removed, the resulting kernel contains the parity–time ($\mathcal{PT}$) core $\Delta_{\rm eff}\sigma_x + i\Gamma_{\rm PT}\sigma_z$ together with its causal Kramers–Kronig detuning. The normalized constant-core benchmark becomes an avoided coalescence when its full finite-band frequency dependence is restored. By contrast, an exactly stationary rotating-drive construction, evaluated with the full radial momentum integral and both helicity cuts, supports a continuous family of nonlinear causal pole exceptional points (EPs), certified by a double-zero condition, local winding, and pole exchange. Zero-temperature complete-basis calculations on four z-shifted $N = 10$ and 12 matrix block-Wilson chains continue a representative EP to $U = 0.0025$. On one common parameter contour both chains have the same winding and pole exchange, and a Rouché ratio $0.678 < 1$ preserves the enclosed zero count between them. This is a controlled finite-chain result, not a numerical renormalization group (NRG) result in the thermodynamic limit or a strong-coupling theorem. We also show that divergent biorthogonal projectors cancel in complete propagators and in the particle–hole-symmetric finite-U charge resolvent, so they do not generate a universal screening enhancement. Finally, on an equal-velocity branchwise-linearized submanifold, the exact finite-U Anderson contact matrix is rational in a dressed rapidity and $GL(2,\mathbb{C})$ covariant. Its Yang–Baxter structure survives the non-unitary similarity and becomes a unipotent boundary twist at the EP. This integrability statement does not extend to the full curved, frequency-dependent driven kernel.

I. INTRODUCTION

Driven quantum systems provide a versatile route to steady states and dynamical phases that have no equilibrium analogs. When coherent drive, dissipation, and strong correlations coexist, the resulting many-body dynamics can generate fixed points that reflect both microscopic symmetry and emergent non-Hermitian structure. Recent work has shown that minimizing a driven bosonic field within a Keldysh mean-field framework can reproduce renormalization-group (RG)-like fixed points [1–3], pointing to a deeper connection between nonequilibrium stability and non-Hermitian criticality [4, 5].

Exceptional points (EPs) are non-Hermitian degeneracies where eigenvalues and eigenvectors coalesce, playing a central role in such settings. They reshape spectral topology and enable non-reciprocal transport, sensing enhancement, and dynamical control. EPs have been linked to the non-Hermitian skin effect [6], higher-order degeneracies [7, 8], and ultrafast projective protocols in both quantum and classical experiments [9, 10]. Their influence extends beyond spectra: non-Hermitian topology underpins phenomena in nonlinear photonics [11, 12], topological acoustics [13], engineered metamaterials [14]; coherent light–matter coupling and engineered dissipation can stabilize correlated states such as topological Kondo phases [15], and nonlinearities can generate instabilities captured by dynamical mean-field theory [16, 17].

Impurity systems are particularly sensitive to anisotropy and dissipation. Boundary-induced openness can mimic potential scattering [18–20], while nonlinear perturbations modify inelastic processes and impurity coherence. Non-Hermitian phase transitions have recently been observed in condensed-matter experiments [21], and nonlinear optical platforms provide precise control of non-Hermitian topology [22–24]. In correlated impurity models, non-Hermitian features manifest through sign reversals in RG flows and causality violations in both interacting and non-interacting settings [4, 25–27]. Non-Hermitian Kondo physics has been explored via numerical renormalization group, cold-atom platforms, and Floquet-engineered conduction bands [28–30]. Recent work on a non-Hermitian Anderson model with complex hybridization has also established a genuine ground-state Kondo-breakdown transition by self-consistent slave-boson and interacting Bethe-Ansatz methods [31]. The present problem is different: its non-Hermitian structure is a passive retarded kernel derived from a Hermitian driven system, and its RG result is deliberately restricted to a weak-flow pre-EP diagnostic. An exact limiting precedent for an emergent non-Hermitian impurity generator was given by Oguri and Sakano [32]: at infinite bias (or in the corresponding high-temperature limit) the Anderson impurity maps to a finite Liouville–Fock Hamiltonian, while physical Green functions require the complete doubled-space initial/final-state construction. That high-energy limit is distinct from the present adiabatic passive kernel, but it reinforces the need to retain complete retarded, advanced, and lesser objects rather

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than assigning observability to an individual left or right eigenvector residue. Integrability of the Kondo model with time-dependent couplings has been established via quantum Knizhnik–Zamolodchikov equations [33], and a $\mathcal{PT}$-protected Kondo effect has been observed in $CeCo_2P_2$ [15]. Benchmark studies of the two-dimensional Hubbard model have established the regime of validity of large-N slave-boson and mean-field treatments of correlated electrons [34], directly motivating the approach employed in Sec. V.

The exact algebraic question is separated from the microscopic derivation. The present article derives the driven retarded kernel and identifies the controlled low-energy submanifold on which its contact problem reduces to a finite-U Anderson model. A companion mathematical work [35] develops that reduced model independently: it proves the finite-volume gauge equivalence, Yang–Baxter equation (YBE), RLL, and RTT construction, unipotent boundary-twist limit, the resolution of the nonzero transverse pseudospin-ladder sector, the resulting many-body Jordan structure, and the Laguerre-scaled contraction of descendant spin roots at $\lambda = \infty$. This division of labor is important. The exact result below is not a claim that curvature, unequal branch velocities, or the full frequency-dependent self-energy of the driven model are Yang–Baxter integrable.

Here we show that a driven impurity embedded in a Dirac-like bath naturally develops a passive non-Hermitian kernel once higher angular harmonics and the SOC matrix bath are treated together. The off-shell auxiliary harmonics generate a real spin-odd coherent shift. The retarded Dirac propagator, through its energy and spin-orbit matrix structure, converts this shift into a spin-dependent decay rate. After subtracting the common bath damping, the remaining co-decaying kernel contains a dynamically induced passive $\mathcal{PT}$-symmetric core, together with a Kramers–Kronig-related coherent detuning that bounds the approach to exact coalescence. We demonstrate that the resulting finite-drive near-EP structure is compatible with the bath-controlled thermal closure—despite arising without any explicit non-Hermitian term in the microscopic Hamiltonian. The principal spectral figures are normalized snapshots of the compensated constant core; they test its algebra and dynamics but do not establish that the microscopic finite-band drive reaches the benchmark EP. A separate low-temperature $U \to \infty$ saddle verifies projected-channel scaling and charge continuity, while no finite-r branch is found at the reference temperature $T_b = 0.1$; no finite-temperature enhancement of the condensate is claimed. *In contrast to earlier studies where non-Hermiticity is imposed phenomenologically or emerges through renormalization-group flow [26, 27, 36], we demonstrate a complementary microscopic route in which a strictly Hermitian, periodically driven Dirac impurity produces a passive non-Hermitian relative kernel after coarse graining and removal of the common decay.* Dirac baths provide a natural setting because spin–momentum locking, angular-momentum selection rules, and an energy-dependent spectral width make the passive $\mathcal{PT}$ channel symmetry constrained.

The paper is organized as follows. Section II defines the full Hermitian Hamiltonian. Section III performs the bilinear Gaussian coarse graining and establishes the emergent $\mathcal{PT}$ symmetry. Section IV derives the steady-state Hamiltonian and the Hadamard rotation that makes hybridization diagonal. Section V presents the slave-boson mean field and the resulting shifted resonance condition. Sections VI–VII state the spectral and fluctuation–dissipation checks, whose full numerical panels are collected in the Supplemental Material. Section VIII discusses real-time dynamics in the passive co-decaying frame. Section IX derives the Jordan response and full scattering eigenphases, then obtains the exact finite-U contact matrix on the controlled branchwise-linearized submanifold and states its metric-covariant Yang–Baxter structure. Section X separates the analytic frozen estimate from the supporting weak-coupling flow and from any thermodynamic Kondo-temperature claim. Section XI concludes.

II. MODEL AND HAMILTONIAN

The full Hermitian Hamiltonian is $H = H_{\rm bath} + H_{\rm imp} + H_{\rm imp\text{-}bath} + H_O + H_{\rm hyb}[\phi(t)]$. Each term has a distinct physical role. $H_{\rm bath}$ describes the two-dimensional Dirac conduction band with spin-orbit coupling. $H_{\rm imp}$ is the correlated Anderson impurity. $H_{\rm imp\text{-}bath}$ is the bare hybridization between the impurity and the $m = 0$ bath channel. H_O represents auxiliary angular-momentum modes generated by the trigonal crystal anisotropy. $H_{\rm hyb}$ couples the physical bath to the angular harmonics and is modulated by the periodic drive. We define all symbols at first use below. To prevent notation clashes, we reserve $t_{\rm av} = (t+t')/2$ and $\tau = t - t'$ for the Wigner center and relative times, respectively, and use T_b only for the reservoir temperature. The chiral bath index is $\eta = \pm$, whereas $\zeta_\sigma = +1$ (-1) for $\sigma = \uparrow$ $(\downarrow)$. The SOC strength is λ; β_0 is the signed frozen drive-control coordinate; and $\bar{\beta} = |\beta_0||b_c|$ is its nonnegative slave-boson mean-field (SBMF)-renormalized projection. The normalized compensated-core figures use $k = 0$ and contain no physical density of states (DOS) or cutoff. The independent causal and reachability audits use a radial cutoff $k_{\rm max} = \pi/4$ and, for the sharp-cutoff Dirac transform only, the separate ultraviolet energy cutoff $D_{\rm uv} = \pi/4$. We denote the analytic compensated-core EP by $\beta_{0,\rm core}^{\rm EP} = 0.50$. On the path used below, $\Delta_{\rm eff}^{(0)} = 0.10$ and $\Gamma_{\rm PT}^{(0)} = 0.20\beta_0$, so β_0 is not itself a decay rate. At the SOC-degenerate $k = 0$ point the bath-dressed double-root condition is identical to this core condition. The finite-k root search is boundary-limited at the upper edge $\beta_0 = 0.55$ of the declared scan and does not establish a separate interior projected EP.

A. Bath

We consider a two-dimensional (2D) host crystal with trigonal (C_3) anisotropy. The conduction electrons form a Dirac bath with spin-orbit coupling (SOC):

$$H_{\rm bath} = \sum_{k,\sigma} \epsilon_k \, c^\dagger_{k\sigma} c_{k\sigma} + \lambda \sum_k k\big(c^\dagger_{k\uparrow} c_{k\downarrow} + \text{h.c.}\big), \qquad (1)$$

where $c^\dagger_{k\sigma}$ creates a bath electron with radial momentum k (in units of inverse lattice constant) and spin $\sigma \in \{\uparrow, \downarrow\}$; $\epsilon_k = \alpha_{\rm b} k^2$ ($\alpha_{\rm b} > 0$), with radial cutoff $0 \le k \le k_{\rm max} = \pi/4$; and $\lambda > 0$ is the spin-orbit coupling. The separate energy cutoff $D_{\rm uv}$ enters only logarithmic and scaling formulas below. The chiral rotation $c_{k\pm} = (c_{k\uparrow} \pm c_{k\downarrow})/\sqrt{2}$ diagonalizes $H_{\rm bath}$ exactly, yielding SOC-split chiral dispersions $\varepsilon_{k\pm} = \alpha_{\rm b} k^2 \pm \lambda k$ with $\varepsilon_{k+} \neq \varepsilon_{k-}$ for all $k \neq 0$. For the full radial dispersions, the two positive-energy roots give

$$\rho_+(\omega) = \frac{\Delta - \lambda}{4\pi\alpha_{\rm b}\Delta}, \qquad \rho_-(\omega) = \frac{\Delta + \lambda}{4\pi\alpha_{\rm b}\Delta}, \qquad \Delta = \sqrt{\lambda^2 + 4\alpha_{\rm b}\omega}. \qquad (2)$$

whenever both roots lie inside the radial cutoff. Hence $\rho_+ + \rho_- = 1/(2\pi\alpha_{\rm b})$ for $\omega \to 0^+$: the complete quadratic-plus-SOC radial bath is not pseudogapped. The familiar $|\omega|/\lambda^2$ law describes only the small-k cone patch of one branch. Below zero energy the lower branch instead has a band minimum at $-\lambda^2/(4\alpha_{\rm b})$ and the associated radial van Hove edge. The constant *sum* does not eliminate the drive-induced relative decay: the channel difference and the SOC matrix term g_x^2 remain nonzero, as shown explicitly in Eq. (25). The full-k calculation below retains this exact structure; the symmetric linear-DOS form is used only as an explicitly idealized analytic benchmark.

B. Impurity

The correlated impurity is described by the Anderson Hamiltonian:

$$H_{\rm imp} = \sum_\sigma \epsilon_\xi \, d^\dagger_\sigma d_\sigma + U \, n_\uparrow n_\downarrow, \qquad (3)$$

where $d^\dagger_\sigma$ creates an impurity electron with spin σ, $n_\sigma = d^\dagger_\sigma d_\sigma$ is the occupation, ϵ_ξ is the bare impurity level before the projected bath-induced detuning is included, and $U > 0$ is the on-site Coulomb repulsion.

C. Impurity–Bath Hybridization

The impurity couples directly to the $m = 0$ (s-wave) bath channel via

$$H_{\text{imp-bath}} = \sum_{k,\sigma} \Big(W_k \, c^\dagger_{k,0,\sigma} d_\sigma + \text{h.c.} \Big), \qquad (4)$$

where W_k is the bare hybridization amplitude. This term defines the physical coupling between the impurity and the conduction bath before the angular-momentum harmonics are introduced. Importantly, W_k is time independent and is not proportional to the frozen drive control β_0. At the slave-boson saddle the physical spinon–bath vertex is rW_k, with $r = |b_c|$.

In the presence of the driven angular-momentum sector [Eqs. (5)–(6)], the $m = 0$ bath propagator becomes dressed by virtual processes involving the auxiliary modes $m \neq 0$: the auxiliary modes generate a retarded self-energy $\Sigma^{\rm bath}_\sigma$ on the $m = 0$ bath channel (Sec. III). The impurity self-energy then arises from this dressed bath propagator through Eq. (4): it is the auxiliary dressing of the bath propagator, communicated to the impurity via the bare coupling W_k, that produces the effective non-Hermitian self-energy on the impurity sector. This two-step mechanism—auxiliary modes dress the bath, bath dresses the impurity—is made explicit in Sec. III and Supplemental Material.

D. Auxiliary Angular-Momentum Modes

The trigonal anisotropy $\beta_3(k) k^3 \cos 3\theta$ of the host crystal (where $\beta_3(k)$ is the momentum-dependent trigonal crystal-field strength, distinct from the hybridization amplitude β_0 introduced below; and θ is the in-plane momentum angle) couples the $m = 0$ (s-wave) bath channel to higher angular-momentum harmonics $m = \pm 3, \pm 6, \ldots$ via the C_3 selection rule $\Delta m = \pm 3$. These higher harmonics are represented by auxiliary fermions $O_{k,m,\sigma}$ with angular momentum index $m \neq 0$:

$$H_O = \sum_{k,m\neq 0,\sigma} \epsilon^{(m)}_O \, O^\dagger_{k,m,\sigma} O_{k,m,\sigma} + \sum_{k,m,\sigma} \beta_3(k) k^3 O^\dagger_{k,m+3,\sigma} O_{k,m,\sigma} + \text{h.c.}, \qquad (5)$$

where the mode energies satisfy $\epsilon^{(m)}_O = \epsilon^{(-m)}_O$ by mirror/time-reversal symmetry of the C_{3v} crystal field. The fields $O_{k,m,\sigma}$ should be understood as a Feshbach, or Loewdin-partitioned, representation of remote higher-angular-harmonic crystal-field channels rather than as an additional low-energy bath. Their energies include the local orbital or crystal-field splitting of these nonresonant harmonics relative to the active $m = 0$ Dirac channel. In the low-energy limit $\epsilon^{(m)}_O \gg D_{\rm uv}$, the auxiliary modes are therefore far off-shell and can be integrated out exactly within the quadratic sector; their main effect is a principal-value self-energy dressing the $m = 0$ bath propagator, which in turn induces a self-energy in the impurity sector through the bare hybridization Eq. (4) (see Sec. III and Supplemental Material). If an auxiliary harmonic approaches the conduction-band window, the same formalism gives an ordinary on-shell bath damp-

ing contribution; this spin-even damping is not identified with the passive relative $\mathcal{PT}$ channel.

E. Periodic Drive and Hybridization

The periodic drive couples the physical bath ($m = 0$) to the remote higher angular harmonics. We write the Hermitian bath–auxiliary hybridization as

$$H_{\rm hyb}[\phi(t)] = \sum_{k,\theta,\sigma,m\neq 0} \big[V_m(k) e^{i3m\theta} e^{i\zeta_\sigma \phi(t)} \times c^\dagger_{k,\theta,\sigma} O_{k,m,\sigma} + \text{h.c.}\big]. \tag{6}$$

Here $\zeta_\uparrow = +1$ and $\zeta_\downarrow = -1$ label the spin-helicity channel selected by the spin-orbit-split Dirac bath. The normalized frozen-kernel scans use the slow oscillatory protocol $\phi(t) = (\pi/4)\sin(\Omega t)$. In that protocol the passive control is the instantaneous angular velocity $\dot\phi(t_{\rm av}) = (\pi\Omega/4)\cos(\Omega t_{\rm av})$; the numerical parameter β_0 used below should be read as the corresponding effective frozen-kernel amplitude, not as the value of the phase itself. Hermiticity fixes the second vertex to carry the complex-conjugate phase. Therefore the normal Gaussian contraction of a number-conserving auxiliary sector contains phase differences, not phase sums. This is important: the off-shell auxiliary sector does not by itself generate anti-Hermitian gain or loss.

Equivalently, the phase can be removed from Eq. (6) by the time-dependent unitary rotation

$$O_{k,m,\sigma} \to e^{-i\zeta_\sigma \phi(t)} O_{k,m,\sigma}. \tag{7}$$

The vertices then become phase-free, while the auxiliary Hamiltonian acquires the Berry term

$$\epsilon^{(m)}_O \to \epsilon^{(m)}_O - \zeta_\sigma \dot\phi(t). \tag{8}$$

Thus a gapped Hermitian auxiliary sector produces a real spin-odd coherent shift. For the oscillatory protocol this statement is an adiabatic low-energy projection. For the stationary rotating protocol $\phi(t) = \Omega t$, however, Eq. (8) is constant and the rotated quadratic Hamiltonian is exactly time independent; no freezing or gradient expansion in Ω is then required. The passive non-Hermitian component appears only in the second step, when this shifted channel is embedded in the energy-dependent retarded Dirac bath. For the oscillatory Gaussian projection the relevant off-shell hierarchy is $|V|/\epsilon^{(m)}_O \ll 1$ and $\Omega \ll \epsilon^{(m)}_O$; a quasistatic interpretation additionally requires the drive to be slow on the retained impurity-relaxation scale. The reachability audit below shows that the normalized EP benchmark does not satisfy these inequalities simultaneously. Because H_O is *bilinear* in $O_{k,m,\sigma}$, the modes $m \neq 0$ can be integrated out at the Gaussian level without approximation within the quadratic sector; the interaction U enters only through the slave-boson mean field on the impurity level (Sec. IV). Since $\dot\phi(t_{\rm av}) \propto \cos(\Omega t_{\rm av})$, a full cycle sweeps through the two signs of the frozen control. The present construction is an instantaneous adiabatic description of that sweep; nonadiabatic EP traversal and energy pumping are outside its scope. The stationary rotating protocol is used below only to provide a separate microscopic reachability witness.

III. COARSE GRAINING AND EMERGENT $\mathcal{PT}$ SYMMETRY

This section isolates the bilinear coarse graining from the later slave-boson mean-field treatment of the local interaction. The raw retarded self-energy generated by the auxiliary angular modes is, in general, a complex retarded function; it is not assumed to be positive. The $\mathcal{PT}$-active low-energy kernel is obtained by projecting the spin-dependent impurity self-energy onto a spin-conjugate pair z, z^*. This formulation keeps the microscopic origin of the passive relative gain–loss channel explicit while avoiding any assumption that the raw retarded envelope is real, positive, or sign definite.

A. Keldysh Path Integral and Angular-Mode Integration

We work within the Keldysh formalism [1], writing the partition function as a coherent-state path integral on the closed time contour. The bath and auxiliary angular modes enter quadratically and can be eliminated at the Gaussian level. The local Coulomb interaction on the impurity is not treated in this Gaussian step; it is incorporated later through the slave-boson mean-field approximation in Sec. V.

The Grassmann integration can be written explicitly. For fixed (k, θ, σ), collect the auxiliary fields into a vector $O = (\dots, O_m, \dots)^T$ and write the quadratic contour action as

$$S_O + S_{cO} = \int_{\mathcal{C}} dt dt'\, \bar O(t) A_O(t,t') O(t') + \int_{\mathcal{C}} dt\, \left[\bar c(t) V_\sigma(t) O(t) + \bar O(t) V^\dagger_\sigma(t) c(t)\right]. \tag{9}$$

The Gaussian identity for Grassmann variables gives

$$\int \mathcal{D}[\bar O, O] e^{i(S_O + S_{cO})} \propto \det A_O\, e^{iS_{\rm infl}}, \tag{10}$$

$$S_{\rm infl} = -\int_{\mathcal{C}} dt dt'\, \bar c(t)\, \Xi_\sigma(t,t')\, c(t'), \tag{11}$$

$$\Xi_\sigma(t,t') = V_\sigma(t) g_O(t,t') V^\dagger_\sigma(t'). \tag{12}$$

Here $g_O = A_O^{-1}$ is the contour auxiliary propagator. This is the complete Gaussian step: because the auxiliary sector conserves particle number, there is no anomalous OO

contraction. Keeping both vertices of Eq. (6) explicit, the normal auxiliary contraction gives

$$\Sigma_\sigma^{O,R}(t,t') = \sum_{m,m'\neq 0} v_{m\sigma}(t)\, g_{mm'}^{O,R}(t,t')\, v_{m'\sigma}^*(t'), \quad (13)$$

$$v_{m\sigma}(t) = V_m e^{i3m\theta} e^{i\zeta_\sigma \phi(t)}. \quad (14)$$

For the diagonal terms, and also for any Hermitian off-shell auxiliary matrix after summing conjugate paths, the phases enter as $e^{i\zeta_\sigma[\phi(t)-\phi(t')]}$. With $T = t_{\rm av}$ and $\tau = t - t'$, this is $e^{i\zeta_\sigma \tau \dot\phi(T)}$ to first adiabatic order, so the Wigner transform shifts the stationary auxiliary self-energy as $\Sigma_0^{O,R}(\omega) \to \Sigma_0^{O,R}[\omega + \zeta_\sigma \dot\phi(T)]$. Its first-order expansion gives the regular Hermitian principal-value shift in Eq. (15); the explicit transform and sign are given in the Supplemental Material. Equivalently, after the unitary rotation in Eq. (7), the same result is obtained from the spin-dependent level shift in Eq. (8).

To leading order in the slow drive and in the off-shell expansion, the auxiliary-dressed $m = 0$ bath therefore has the form

$$\Sigma_\sigma^{\rm bath,R}(\omega, t_{\rm av}) = \Sigma_0^{\rm bath,R}(\omega) + \zeta_\sigma \Delta_{\rm aux}(\omega, t_{\rm av}) + O\big[(\Omega/\epsilon_O)^2\big]. \quad (15)$$

Here $\Delta_{\rm aux}$ is real in the low-energy window. Microscopically one may write, schematically,

$$\Delta_{\rm aux}(\omega, t_{\rm av}) = -\,\dot\phi(t_{\rm av})\, \partial_{\epsilon_O} \sum_{m\neq 0} |V_m|^2 \times \mathcal{P}\frac{1}{\omega - \epsilon_O^{(m)}}, \quad (16)$$

up to form-factor and angular-overlap coefficients. If an auxiliary pole enters the conduction window, the corresponding delta-function part gives ordinary bath damping; it is not identified with the passive $\mathcal{PT}$ channel.

The bare impurity–bath hybridization W_k in Eq. (4) remains time independent. The drive reaches the impurity through the dressed bath propagator. In contour notation the two Gaussian eliminations give

$$G_{c,\sigma}^{-1} = g_c^{-1} - \Xi_\sigma, \qquad \Xi_\sigma = V_\sigma \circ g_O \circ V_\sigma^\dagger, \quad (17)$$

$$\Sigma_{d,\sigma} = W^\dagger \circ G_{c,\sigma} \circ W, \quad (18)$$

where $\circ$ denotes contour convolution. Thus the drive-dependent virtual path is $d \to c \to O \to c \to d$; no direct time dependence of W_k is required. The full Schur-complement derivation is given in the Supplemental Material.

In the frozen off-shell window, write $\Xi_\sigma^R = \Xi_0^R + \zeta_\sigma \delta\Xi^R$ with $\delta\Xi^R \in \mathbb{R}$. Expanding the dressed-bath resolvent gives the general first-order correction

$$\delta\Sigma_d^R = W^\dagger G_{c,0}^R\, \delta\Xi^R\, G_{c,0}^R W. \quad (19)$$

Although the auxiliary correction is real, the retarded continuum propagators are complex, so the transmitted impurity correction has both coherent and dissipative parts. Defining

$$\Sigma_{d,0}^R(\omega) = \Lambda(\omega) - i\bar\Gamma(\omega), \quad (20)$$

we write

$$\Sigma_{d,\sigma}^R = \Sigma_{d,0}^R + \zeta_\sigma \Delta_{\rm coh} + i\zeta_\sigma \Gamma_{\rm PT}. \quad (21)$$

For the nearly rigid scalar shift used in the analytic estimates, $\delta\Xi^R \simeq \Delta_{\rm aux}$ and the spin-even auxiliary background varies slowly. Then

$$\Sigma_{d,\sigma}^R(\omega, t_{\rm av}) \simeq \Sigma_{d,0}^R(\omega - \zeta_\sigma \Delta_{\rm aux}) \simeq \Sigma_{d,0}^R - \zeta_\sigma \Delta_{\rm aux} \partial_\omega \Sigma_{d,0}^R, \quad (22)$$

which gives

$$\Delta_{\rm coh} = -\Delta_{\rm aux} \partial_\omega \Lambda, \qquad \Gamma_{\rm PT} = \Delta_{\rm aux} \partial_\omega \bar\Gamma. \quad (23)$$

Equation (19) is the general first-order result; Eq. (22) is its rigid-shift limit. All functions are evaluated in the active low-energy window. For an SOC matrix bath, however, the exact result need not reduce to the rigid shift of the spin-even scalar self-energy. In the chiral basis set $a = (\omega^+ - \varepsilon_{k+})^{-1}$, $b = (\omega^+ - \varepsilon_{k-})^{-1}$, so that $g_0 = (a+b)/2$ and $g_x = (a-b)/2$. Then

$$-\partial_\omega g_0 = g_0^2 + g_x^2, \qquad g_b \sigma_z g_b = (g_0^2 - g_x^2)\sigma_z = ab\,\sigma_z. \quad (24)$$

With the sign convention of Eq. (16), the exact first-order relative coefficient is therefore

$$\delta\Sigma_z^R = \Delta_{\rm aux} \sum_k |W_k|^2 a_k b_k,$$

$$\Gamma_{\rm PT} = \Delta_{\rm aux}\left[\partial_\omega \bar\Gamma - 2\,{\rm Im}\sum_k |W_k|^2 g_{x,k}^2\right]. \quad (25)$$

Reversing the definition of the spin-odd auxiliary shift reverses both sides of Eq. (25), without changing any rate eigenvalue. Consequently a structureless scalar flat band ($g_x = 0$) cannot generate the relative decay at this order, whereas an SOC-split bath can do so even when its *total* width is exactly flat: for $\partial_\omega \bar\Gamma = 0$ the decay imbalance is carried entirely by the g_x^2 term. This is the operative mechanism in the full-momentum certificate below. The SOC split bath supplies the spin–momentum-locked helicity channels, and in that basis

$$\bar\Gamma_\eta(\omega) = \pi \sum_k |W_{k\eta}|^2 \delta(\omega - \varepsilon_{c,\eta}(k)), \quad (26)$$

$$\varepsilon_{c,\eta}(k) = \alpha_{\rm b} k^2 + \eta\lambda k. \quad (27)$$

so the SOC parameter λ controls the channel-resolved densities of states and the matrix form factor entering the projected EP diagnostics. Although their positive-energy sum is flat in the full radial model, their difference and g_x^2 are not. Without SOC-induced helicity structure the auxiliary shift remains a coherent real field rather than a passive relative decay channel in this flat-total-width realization.

B. Passive $\mathcal{PT}$ Core and Causal Detuning

The projected impurity self-energy is decomposed into a common bath part, a passive relative decay imbalance, and a Kramers–Kronig-related coherent detuning:

$$\begin{aligned}\Sigma^R_{\rm imp}(\omega,t_{\rm av}) &= \left[\Lambda(\omega)-i\bar{\Gamma}(\omega)\right]\mathbb{I}\\ &\quad+\Delta_{\rm coh}(\omega,t_{\rm av})\sigma_z+i\Gamma_{\rm PT}(\omega,t_{\rm av})\sigma_z.\end{aligned}\tag{28}$$

The first line is a common Lamb shift and common decay. It is retained in the bath-controlled FDR, but it does not affect the relative EP structure. The second line contains two spin-odd terms with different symmetry status: $i\Gamma_{\rm PT}\sigma_z$ is the passive relative decay imbalance, while $\Delta_{\rm coh}\sigma_z$ is a real coherent detuning generated by the same causal retarded bath self-energy.

The SOC bath supplies the Hermitian spin-mixing channel. In the original spin basis, the local bath Green function has the matrix structure

$$\begin{aligned}g^R_b(k,\omega) &= \left[\omega^+-\epsilon_k-\lambda k\sigma_x\right]^{-1}\\ &= g_0(k,\omega)\mathbb{I}+g_x(k,\omega)\sigma_x,\end{aligned}\tag{29}$$

with

$$g_x(k,\omega)=\frac{\lambda k}{(\omega^+-\epsilon_k)^2-\lambda^2k^2}.\tag{30}$$

Equivalently, with $\epsilon_{k\pm}=\epsilon_k\pm\lambda k$,

$$\begin{aligned}g_x &= \frac{1}{2}\left[\frac{1}{\omega^+-\epsilon_{k+}}-\frac{1}{\omega^+-\epsilon_{k-}}\right],\\ \Sigma_x &= \frac{1}{2}(\Sigma_+-\Sigma_-)=\frac{\Lambda_+-\Lambda_-}{2}-i\frac{\Gamma_+-\Gamma_-}{2}.\end{aligned}\tag{31}$$

The principal-value part of this SOC off-diagonal bath dressing gives a real coherent spin-mixing coefficient,

$$\Delta_{\rm eff}(\omega)=\sum_k|W_k|^2\,{\rm Re}\,g_x(k,\omega),\tag{32}$$

renormalized in the interacting problem by the slave-boson saddle point. Equation (32) names only that coherent principal-value component in the normalized Markov core. The full causal kernel retains the imaginary part in Eq. (31) together with its complete frequency dependence. The drive-induced shift cannot leak into this mixing channel at leading order: expanding the dressed bath propagator to first order in the spin-odd shift $\Delta_{\rm aux}\sigma_z$ gives

$$g^R_b\,\sigma_z\,g^R_b=\left(g_0^2-g_x^2\right)\sigma_z,\tag{33}$$

because $\{\sigma_x,\sigma_z\}=0$, so the first-order correction is purely σ_z and $\Delta_{\rm eff}$ remains a static (drive-independent) channel at this order, corresponding to the offset Δ_0 of Eq. (48); the isolated finite-drive EP of the generic projection is therefore the microscopically natural case. Thus, after subtracting the common damping, the general relative two-channel kernel is

$$h_{\rm rel}=\Delta_{\rm eff}\sigma_x+\left(\Delta_{\rm coh}+i\Gamma_{\rm PT}\right)\sigma_z.\tag{34}$$

The passive-$\mathcal{PT}$-symmetric core is obtained at the compensated detuning point $\Delta_{\rm coh}=0$:

$$h_{\mathcal{PT}}=\Delta_{\rm eff}\sigma_x+i\Gamma_{\rm PT}\sigma_z,\qquad \sigma_x h^*_{\mathcal{PT}}\sigma_x=h_{\mathcal{PT}}.\tag{35}$$

The real term $\Delta_{\rm coh}\sigma_z$ is not discarded. It is $\mathcal{PT}$-odd under the same operation and therefore acts as a causal detuning floor. The pole splitting of Eq. (34) is

$$s^2=\Delta^2_{\rm eff}+\left(\Delta_{\rm coh}+i\Gamma_{\rm PT}\right)^2.\tag{36}$$

For real $\Delta_{\rm eff}$, $\Delta_{\rm coh}$, and $\Gamma_{\rm PT}$, exact coalescence in the passive sector requires

$$\Delta_{\rm coh}\Gamma_{\rm PT}=0,\qquad \Delta^2_{\rm eff}+\Delta^2_{\rm coh}-\Gamma^2_{\rm PT}=0.\tag{37}$$

Thus, for a nonzero relative decay imbalance, the exact passive EP occurs at the compensated point $\Delta_{\rm coh}=0$ and $|\Delta_{\rm eff}|=|\Gamma_{\rm PT}|$. In the idealized scalar cone-patch estimate, compensation is a codimension-one operating-energy condition: $\Delta_{\rm coh}\propto\partial_\omega\Lambda$ changes sign and vanishes at a finite energy of order $D_{\rm uv}/e$ for the linear Dirac width, where $\partial_\omega\bar{\Gamma}$ remains finite, while near the Fermi level the companion ratio is logarithmically enhanced, $|\Delta_{\rm coh}/\Gamma_{\rm PT}|\simeq(2/\pi)\ln(D_{\rm uv}/|\omega|)$. Away from the compensated point, the Kramers–Kronig detuning prevents a literal pole coalescence. Minimizing $|s|$ over real $\Delta_{\rm eff}$ at fixed $\Delta_{\rm coh}$ and $\Gamma_{\rm PT}$ gives

$$|s|_{\rm min}=\begin{cases}\sqrt{2|\Delta_{\rm coh}\Gamma_{\rm PT}|}, & |\Delta_{\rm coh}|\le|\Gamma_{\rm PT}|,\\ \sqrt{\Delta^2_{\rm coh}+\Gamma^2_{\rm PT}}, & |\Delta_{\rm coh}|>|\Gamma_{\rm PT}|,\end{cases}\tag{38}$$

with the square-root non-analyticity characteristic of a second-order EP. This causal detuning should not be identified with a universal interaction-induced splitting. In the exact equal-velocity, scalar-hybridization linearized reduction derived below, the standard local interaction preserves the complexified pseudospin symmetry and does not lift the EP [35]. Curvature, unequal velocities, an energy-dependent self-energy, or component-dependent vertices can instead move or split it. The Supplemental Material derives the minimization and the *idealized symmetric-cone* sharp-cutoff Dirac Kramers–Kronig estimate explicitly. For that benchmark, $D_{\rm uv}=\pi/4$ gives an exact compensation energy $|\omega_c|=0.24608$ (the low-energy estimate is $D_{\rm uv}/e=0.28893$). Compensation holds only at ω_c: one common half-width $\bar{\Gamma}=0.12$ away, the residual splitting floor is already 0.90–1.00 times the relative channel in the audited benchmark. The full finite-band causal kernel therefore exhibits an avoided coalescence over a linewidth even though the constant compensated core is exactly defective.

This two-step structure is the microscopic origin of the kernel used in the rest of the paper: off-shell auxiliary modes generate a real spin-odd shift, and the retarded SOC matrix bath converts that shift into a spin-dependent width while also supplying the Hermitian spin-mixing channel. The numerical scans treat β_0

as a tunable frozen effective kernel control parameter. In the scalar cone-patch rigid-shift estimate,

$$\Gamma_{\rm PT} \sim \Omega\, \frac{|V|^2}{\epsilon_O^2}\, \partial_\omega \bar{\Gamma}, \tag{39}$$

whereas the full radial model uses the additional SOC term in Eq. (25). Thus the construction establishes the symmetry-allowed origin of the passive relative channel, but does not determine the numerical magnitude of the dimensionless frozen-kernel control β_0. More explicitly, writing $x = |V|/\epsilon_O$ and using the phase amplitude $\pi/4$, the benchmark $\Gamma_{\rm PT}/\bar{\Gamma} = 0.10/0.12$ requires approximately

$$\frac{\Omega}{\epsilon_O} \gtrsim \frac{1.061}{x^2}. \tag{40}$$

This is incompatible with both $x \ll 1$ and a quasistatic oscillatory drive. The passivity condition itself is satisfied ($\Gamma_{\rm PT} = 0.10 < \bar{\Gamma} = 0.12$), but microscopic reachability of the normalized EP is not established. Figure 2 therefore exposes the exact constant-core geometry only. The independent causal test in Supplemental Fig. S5 uses the same small width relative to the unit charge gap and quantifies the finite-band avoided-coalescence correction.

a. Stationary full-momentum witness. The preceding negative statement is specific to the normalized benchmark and the oscillatory quasistatic protocol. For $\phi(t) = \Omega t$, the rotating-frame auxiliary levels are shifted by the constant $\mp\Omega$. Keeping their exact frequency dependence, the curved SOC dispersion, and the complete radial momentum integral, define on the outgoing resonance sheet

$$F(z; \epsilon_d, V_O) = \det\left[(z - \epsilon_d)\mathbb{I} - \Sigma_d^{\rm II}(z; V_O)\right]. \tag{41}$$

A nonlinear pole EP is a double zero $F = \partial_z F = 0$ with $\partial_z^2 F \neq 0$ and a rank-one inverse kernel. To certify its existence without evaluating a singular eigenbasis, let $z_c(\epsilon_d, V_O)$ be the continuously branch-tracked solution of $\partial_z F(z_c) = 0$ on a closed control loop C, and define

$$\mathcal{D}_{\rm loc} = F(z_c; \epsilon_d, V_O), \qquad W_C = \frac{1}{2\pi} \Delta_C \arg \mathcal{D}_{\rm loc}. \tag{42}$$

When $\partial_z^2 F$ stays nonzero, the implicit-function theorem keeps z_c on one smooth critical-point branch. Regarding $\mathcal{D}_{\rm loc}$ as a map from the two real controls to $(\mathrm{Re}\,\mathcal{D}_{\rm loc}, \mathrm{Im}\,\mathcal{D}_{\rm loc})$, its boundary winding is the Brouwer degree whenever $\mathcal{D}_{\rm loc} \neq 0$ on C. A nonzero degree forces at least one enclosed zero (with regular zeros counted by their Jacobian sign). At such a zero, $F = \partial_z F = 0$; the nonzero second derivative makes it a quadratic pole coalescence, and the rank-one inverse-kernel test excludes a scalar degeneracy. Explicit pole exchange and the half-winding of their separation provide the equivalent EP2 monodromy check. The full two-cut calculation in the Supplemental Material gives such a point for the better-conditioned representative point

$$\begin{gathered} \lambda = 0.05, \quad W = 0.10, \quad k_{\max} = \pi/4, \\ \epsilon_O = 120, \quad \Omega = 60, \end{gathered} \tag{43}$$

at

$$\begin{aligned} V_O &= 1.21342201, \qquad \epsilon_d = 0.01517035, \\ z_{\rm EP} &= 0.01197283 - 0.00280722i. \end{aligned} \tag{44}$$

Both helicity cuts are open. The driven $k = 0$ upper threshold is $\omega_{\rm th}^{+} = 0.00817959$, so the pole lies 0.00379325, or $1.351|\,\mathrm{Im}\, z_{\rm EP}|$, above it; this threshold proximity and the lower-branch van Hove structure are included in the analytic continuation. Numerically, $|F| = 1.44 \times 10^{-21}$, $|\partial_z F| = 1.09 \times 10^{-15}$, and $\partial_z^2 F = 1.68645 - 0.19034i$; the local degree is $+1$ and the two poles exchange on a closed loop in (ϵ_d, V_O). The physical rate-matrix eigenvalues are 1.904×10^{-3} and 3.096×10^{-3}, with $\bar{\Gamma} = 0.002500$ and passivity margin $\Gamma_{\rm min}/\bar{\Gamma} = 0.761$. Moreover, $V_O/(\epsilon_O - \Omega) = 0.02022$, while the empty-auxiliary thermal exponent is $(\epsilon_O - \Omega)/T_{\rm b} = 600$ at $T_{\rm b} = 0.1$. Thus $\Omega/\bar{\Gamma} \simeq 2.4 \times 10^4$: the oscillatory quasistatic condition is strongly violated, whereas no such condition is required because the rotating-frame Hamiltonian is exactly stationary. The EP certificate is a retarded-kernel statement and is temperature independent. At the displayed temperature $T_{\rm b}/\bar{\Gamma} = 40$, so linewidth-scale spectral or transient signatures are thermally unresolved; observing them would instead require $T_{\rm b} \lesssim 10^{-3}$ for this witness.

Continuation in SOC, with W held fixed at 0.10 while the real gate and the off-shell auxiliary vertex are retuned by the same double-zero equations, gives a finite locus over $0.02 \leq \lambda \leq 0.1578$. Independent local-winding and pole-exchange certificates pass at $\lambda = 0.05$, 0.10, and 0.15. The continued pole approaches an open-channel threshold near the upper end while the rate matrix remains passive; this threshold termination is not a passivity breakdown. No endpoint or EP phase boundary is claimed. This establishes a controlled microscopic family of causal pole EPs in the full curved kernel. It does not turn the normalized benchmark into a causal EP, identify the nonlinear pole EP with the nilpotent Markov-core EP, or imply full-momentum Yang–Baxter integrability.

b. Finite-U block-Wilson continuation. We next test whether the representative $\lambda = 0.05$ causal pole EP survives a local Anderson interaction without imposing the double-zero condition on the interacting self-energy. The positive 2×2 matrix spectral density of the same rotating full-k bath is logarithmically discretized with $\Lambda = 1.5$, mapped by block Lanczos to a matrix Wilson chain, and averaged over four z shifts. We first use 12-site chains to locate the finite-U center and evaluate the zero-temperature correction-vector Green functions G and F in exact fixed-charge sectors and form the equation-of-motion self-energy

$$\Sigma_U(z) = U F(z) G^{-1}(z). \tag{45}$$

Solving $F_U = \partial_z F_U = 0$ for the determinant of the full analytic two-cut kernel dressed by Eq. (45) gives, at $U =$

0.0025,

$$z_{\rm EP}^{(U)} = 0.01177670 - 0.00225815i, \\ \epsilon_d^{(U)} = 0.01440607, \qquad W^{(U)} = 0.09036305. \tag{46}$$

Here $U/D_{\rm uv} = 3.18 \times 10^{-3}$, while $U/\bar{\Gamma}_{\rm self} = 1.225$ with $\bar{\Gamma}_{\rm self} = (W^{(U)})^2/4$. Thus the interaction is small on the bandwidth scale but moderate on the retuned local-width scale; it is not a deep-Kondo or strong-coupling calculation.

Figure 1 gives the independent topological gate. On an ellipse with half-widths 10^{-4} in ϵ_d and 10^{-3} in W, the branch-tracked local discriminant has winding -1, the two poles exchange, and their separation has half-winding $-1/2$. The quadratic, phase-step, pole-residual, and branch-assignment guards all pass (see the Supplemental Material). Increasing the z-average from four to eight shifts changes the four fitted coordinates by at most 0.71%, while the exact sum rule and equation-of-motion identity hold to roundoff. A kept-space calculation at $N_{\rm keep} = 960$ differs from the exact finite-chain response by at most 4.18% in the self-energy and 1.40% in the sum rule. These are convergence diagnostics; the complete-basis response carries the topology claim. No infinite-chain, finite-temperature full-density-matrix NRG (FDM-NRG), phase-boundary, or universal interaction-scaling conclusion is drawn.

As an independent chain-length gate, the center is reoptimized separately on the $N = 10$ and $N = 12$ complete-basis chains. Their maximum relative coordinate drift is 3.746×10^{-3}. On a shared ellipse with half-widths 2×10^{-4} in ϵ_d and 2×10^{-3} in W, both chains have local-discriminant winding -1, endpoint pole exchange, and pole-separation half-winding $-1/2$. At 256 contour points,

$$\frac{\sup_C |D_{12} - D_{10}|}{\min_C |D_{10}|} = \frac{1.638 \times 10^{-9}}{2.417 \times 10^{-9}} = 0.6775 < 1, \tag{47}$$

so Rouché's theorem preserves the enclosed zero count between the two finite chains: the boundary deformation from D_{10} to D_{12} cannot cross the origin. Here D_N is the finite-chain version of Eq. (42), with its critical pole branch tracked continuously on the shared contour. The same gate passes at 128 points. This establishes finite-N persistence over the tested pair, not an $N \to \infty$ limit.

IV. STEADY-STATE HAMILTONIAN AND EMERGENT NON-HERMITICITY

Given the passive spin-conjugate relative self-energy from Sec. III, this section derives the explicit 4×4 steady-state Hamiltonian $\tilde{H}(k)$ and identifies the structural origin of its exceptional points. A key step is the Hadamard rotation that diagonalizes the bare hybridization block, revealing that each chiral spinon mode couples directly to one bath channel. The internal $i\Gamma_{\rm PT}$ term still mixes those impurity modes, so the physical two-channel scattering matrix need not be diagonal; the frozen finite-size quantization below is therefore formulated using its exact eigenphases.

A. From Self-Energy to Steady-State Hamiltonian

After the Gaussian coarse-graining of the bath and auxiliary angular modes in Sec. III, and after introducing the slave-boson saddle point in Sec. V, the retarded spinon action takes the form $S^R_{\rm imp} = \sum_k \bar{f}[\omega\mathbb{I} - H_{\rm ss}(k)]f$, where the steady-state Hamiltonian $H_{\rm ss}(k)$ is defined as the inverse of the retarded Green's function at the saddle point: $H_{\rm ss}(k) \equiv [G^R_{\rm ss}(\omega,k)]^{-1}|_{\rm saddle}$. The gauge-invariant slave-boson amplitude is $r = |b_c|$. The three channels in the effective Hamiltonian have distinct microscopic origins: the passive relative width and coherent detuning are components of the projected retarded self-energy, whereas the impurity–bath vertex descends from the time-independent bare coupling W. Since an impurity self-energy contains two spinon–bath vertices, the physical projected parametrization is

$$\Gamma_{\rm PT} = r^2\Gamma^{(0)}_{\rm PT}(\beta_0), \qquad \Delta_{\rm eff} = r^2\Delta_0^{(0)}, \\ V_{\rm eff} = rW. \tag{48}$$

Here the superscript (0) denotes the frozen bath projection before slave-boson renormalization. The core discriminant consequently obeys

$$s^2_{\rm core} = r^4\left\{\left[\Delta_0^{(0)}\right]^2 - \left[\Gamma^{(0)}_{\rm PT}(\beta_0)\right]^2\right\}. \tag{49}$$

For $r > 0$, the common factor r^4 cancels from the core EP condition. The local core EP is therefore exactly invariant under changes of r. A bath-dressed finite-k minimum must be established independently; in the declared scan it is boundary-limited and is not classified as an EP. Figure 2 uses $r = 1$ and a normalized frozen-kernel geometry path; they are intended to display the bifurcation rather than to fix an Anderson hybridization width. The supporting channel-resolved RG calculation uses the conservative fixed-W projection

$$\Gamma_0 = \pi\rho W^2 = 0.12, \\ \Delta^{(0)}_{\rm eff} = 0.10, \qquad \Gamma^{(0)}_{\rm PT} = 0.20\beta_0, \tag{50}$$

for which $\Gamma_0/\Delta_{\rm charge} = 0.12$ and the core EP remains at $\beta_0 = 0.50$. Thus the RG test is performed in a local-moment-compatible small-width regime, with W independent of the drive coordinate.

At the compensated passive-$\mathcal{PT}$ point $\Delta_{\rm coh} = 0$, substituting the projected self-energy into the retarded Dyson equation and including the hybridization to the chiral bath channels gives, in the spin basis

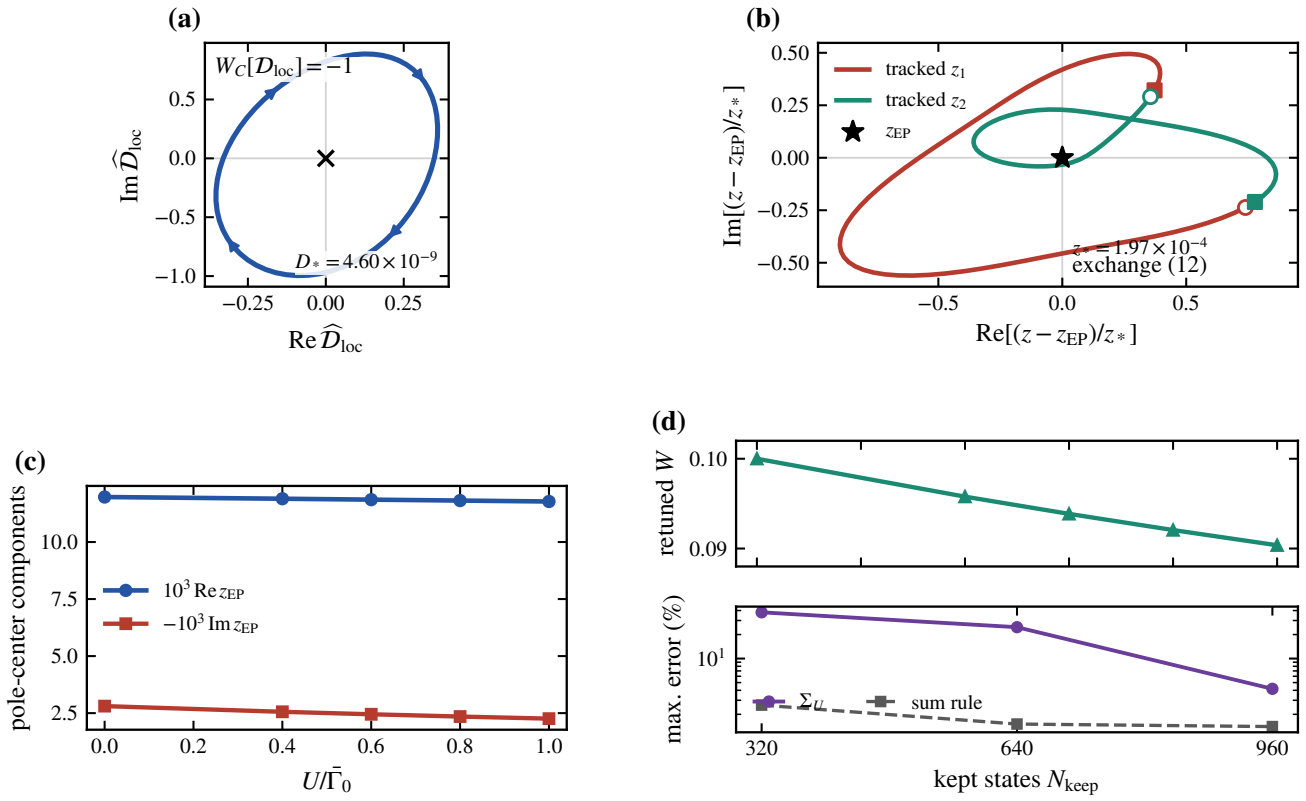


FIG. 1. **Finite-U continuation of the causal full-k pole EP.** (a) The dimensionless contour $\widehat{\mathcal{D}}_{\rm loc} = \mathcal{D}_{\rm loc}/D_*$, with $D_* = \max_C |\mathcal{D}_{\rm loc}| = 4.603 \times 10^{-9}$, winds once clockwise around the origin on the interacting control ellipse at $U = 0.0025$. The positive rescaling removes a plotting unit and cannot change the Brouwer degree. (b) The two nearby poles are plotted as $(z - z_{\rm EP})/z_*$, with $z_* = \max_C |z - z_{\rm EP}| = 1.965 \times 10^{-4}$, and exchange after one circuit; open circles and filled squares mark the initial and final tracked labels, and the star marks Eq. (46). (c) The exact finite-chain center varies smoothly from $U = 0$ to $U/\bar{\Gamma}_0 = 1$, where $\bar{\Gamma}_0 = 0.0025$. (d) The upper panel shows the retuned impurity hybridization; the lower panel compares kept-space results with the exact response on the same four z-shifted 12-site chains. These errors are convergence diagnostics and are not used to establish the winding. The separate $N = 10 \to 12$ common-contour gate is given by Eq. (47) and detailed in the Supplemental Material.

$(d_\uparrow, d_\downarrow, c_{k+}, c_{k-})$,

$$H_{\rm ss}(k) = \begin{pmatrix} \tilde{\epsilon}_\xi + i\Gamma_{\rm PT} & \Delta_{\rm eff} & V_{\rm eff}/\sqrt{2} & V_{\rm eff}/\sqrt{2} \\ \Delta_{\rm eff} & \tilde{\epsilon}_\xi - i\Gamma_{\rm PT} & V_{\rm eff}/\sqrt{2} & -V_{\rm eff}/\sqrt{2} \\ V_{\rm eff}/\sqrt{2} & V_{\rm eff}/\sqrt{2} & \varepsilon_{c,+}(k) & 0 \\ V_{\rm eff}/\sqrt{2} & -V_{\rm eff}/\sqrt{2} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \tag{51}$$

Here $\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma$ is the renormalized impurity level. The Lagrange multiplier δ_c enforces the slave-boson constraint, while δ_Σ is the real projected one-body shift from the principal-value self-energy in Sec. III B (and may be absorbed into ϵ_ξ by convention). The chiral bath dispersions are $\varepsilon_{c,\pm}(k) = \alpha_{\rm b}k^2 \pm \lambda k$. The interaction U is not an explicit two-body coupling in the 4×4 Gaussian kernel; it enters through the SBMF constraint and through the Schrieffer–Wolff and finite-U contact virtual-charge energies used below. For the finite-U Anderson interpretation, particle-hole symmetry is imposed by the common impurity-level choice $\epsilon_\xi = -U/2$, not by adding opposite $\pm U/2$ spin splittings. A real Zeeman field, when present, would enter as an independent spin splitting $B_{\rm imp}$ rather than as the Hubbard interaction.

The $\mathcal{PT}$ symmetry of $H_{\rm ss}(k)$ is explicit for real $\Gamma_{\rm PT}$, $\Delta_{\rm eff}$, and $V_{\rm eff}$: under $P : \uparrow \leftrightarrow \downarrow$ combined with complex conjugation, $(\tilde{\epsilon}_\xi + i\Gamma_{\rm PT}) \to (\tilde{\epsilon}_\xi - i\Gamma_{\rm PT})^* = \tilde{\epsilon}_\xi + i\Gamma_{\rm PT}$, while the real off-diagonal and hybridization entries are exchanged into themselves. Thus the $\mathcal{PT}$ structure is fixed by the projected self-energy channels and the gauge-invariant amplitude $r = |b_c|$, not by the phase of the slave-boson condensate.

B. Diagonal-Hybridization Rotation

The hybridization block of $H_{\rm ss}$ has Hadamard structure $(V_{\rm eff}/\sqrt{2})\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$. The unitary rotation $U_2 = \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]$ acting on the impurity subspace defines chiral spinon modes $\tilde{d}_\pm = (d_\uparrow \pm d_\downarrow)/\sqrt{2}$ and diagonalizes

the hybridization block exactly:

$$U_2^{-1} h_{\rm imp} U_2 = \begin{pmatrix} \tilde{\epsilon}_\xi + \Delta_{\rm eff} & i\Gamma_{\rm PT} \\ i\Gamma_{\rm PT} & \tilde{\epsilon}_\xi - \Delta_{\rm eff} \end{pmatrix}, \tag{52}$$

$$U_2^{-1} V_{dc} = V_{\rm eff}\, \mathbb{I}, \tag{53}$$

where $h_{\rm imp} = \left(\begin{smallmatrix} \tilde{\epsilon}_\xi + i\Gamma_{\rm PT} & \Delta_{\rm eff} \\ \Delta_{\rm eff} & \tilde{\epsilon}_\xi - i\Gamma_{\rm PT} \end{smallmatrix}\right)$ and $V_{dc} = (V_{\rm eff}/\sqrt{2})\left(\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. The algebra is identical to the symmetric case: $\frac{V_{\rm eff}}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\left[\begin{smallmatrix} 1 & 1 \\ 1 & -1 \end{smallmatrix}\right]^2 = \frac{V_{\rm eff}}{2} \cdot 2\mathbb{I} = V_{\rm eff}\mathbb{I}$. The full rotated Hamiltonian in the $(\tilde{d}_+, \tilde{d}_-, c_{k+}, c_{k-})$ basis is:

$$\tilde{H}(k) = \begin{pmatrix} \tilde{\epsilon}_\xi + \Delta_{\rm eff} & i\Gamma_{\rm PT} & V_{\rm eff} & 0 \\ i\Gamma_{\rm PT} & \tilde{\epsilon}_\xi - \Delta_{\rm eff} & 0 & V_{\rm eff} \\ V_{\rm eff} & 0 & \varepsilon_{c,+}(k) & 0 \\ 0 & V_{\rm eff} & 0 & \varepsilon_{c,-}(k) \end{pmatrix}. \tag{54}$$

Here $\tilde{d}_+$ ($\tilde{d}_-$) hybridizes only with c_{k+} (c_{k-}), while the internal $i\Gamma_{\rm PT}$ term can still convert the two asymptotic channels. The frozen impurity eigenvalues are $E_\pm = \tilde{\epsilon}_\xi \pm s$, $s = \sqrt{\Delta_{\rm eff}^2 - \Gamma_{\rm PT}^2}$; on the audited path $\Delta_{\rm eff} = r^2 D$, $\Gamma_{\rm PT} = r^2 G$, $D = 0.10$, and $G = 0.20\beta_0$. The plotted eigenvalues and condition numbers nevertheless use the full 4×4 kernel in Eq. (54). Eliminating its bath block gives

$$\mathcal{P}(z,k) = \mathfrak{p}_+(z,k)\mathfrak{p}_-(z,k) + \Gamma_{\rm PT}^2 = 0, \tag{55}$$

where

$$\mathfrak{p}_\eta(z,k) = z - \tilde{\epsilon}_\xi - \eta\Delta_{\rm eff} - \frac{V_{\rm eff}^2}{z - \varepsilon_{c,\eta}(k)}, \qquad \eta = \pm. \tag{56}$$

The full quadratic-kernel EP condition is

$$\mathcal{P}(z_{\rm EP}, k) = 0, \qquad \partial_z \mathcal{P}(z_{\rm EP}, k) = 0. \tag{57}$$

At $k = 0$ the bath energies coincide, so with $\zeta(z) = z - \tilde{\epsilon}_\xi - V_{\rm eff}^2/z$ this reduces to

$$\mathcal{P}(z,0) = \zeta(z)^2 - \left(\Delta_{\rm eff}^2 - \Gamma_{\rm PT}^2\right). \tag{58}$$

Thus the impurity-related double root occurs at $\Delta_{\rm eff}^2 = \Gamma_{\rm PT}^2$ in this common-shift limit; SOC-resolved finite-k asymmetry generally unfolds it. The declared nonzero-k scan finds no additional interior EP because every minimizer lies at its upper boundary.

At $\beta_0 = 0$, $G = 0$ but $D = 0.10$, so the origin is a normal, coherently split impurity doublet and the full $k = 0$ eigenvector condition number is one. The finite-drive local EP is at $|G| = |D|$, hence $|\beta_0| = 0.50$. In the no-flip control, a normal-matrix guard fixes the arbitrary eigenbasis at the ordinary origin degeneracy and prevents a false condition-number spike.

The operator-resolved finite-U scalar-chain calculation is a distinct Markov-core residue audit: it preserves the nilpotent direction in its validated early-iteration window but supports no universal splitting exponent or quartic coefficient (Supplemental Fig. S7). The interacting continuation of the nonlinear full-k pole EP is instead the matrix block-Wilson complete-basis calculation in Fig. 1; the two are not identified. The scalar local interaction protects the EP only on the equal-velocity, scalar-hybridization linearized submanifold; deviations in the complete driven problem must be evaluated explicitly.

V. SLAVE-BOSON MEAN FIELD AND KELDYSH SELF-CONSISTENCY

The effective impurity problem originates from a correlated Anderson level with local repulsion U. We formulate the constrained low-energy sector using the $U \to \infty$ large-N slave-boson mean-field saddle [30, 37, 38]. The finite-$U = 2$ Schrieffer–Wolff formulas used later are a separate frozen analytic estimate. This section introduces the spinon operators $f_\sigma^\dagger$ that replace the physical $d_\sigma^\dagger$ after the slave-boson decoupling, derives the self-consistent equations for the condensate amplitude $r = |b_c|$ and Lagrange multiplier δ_c, and formulates the shifted resonance condition in gauge-invariant language.

A. Slave-Boson Representation

To handle the Coulomb repulsion U in the Keldysh framework, we use the large-N slave-boson (SBMF) representation [30, 37, 38]:

$$d_\sigma^\dagger = f_\sigma^\dagger b, \qquad b^\dagger b + \sum_\sigma f_\sigma^\dagger f_\sigma = Q. \tag{59}$$

Here $f_\sigma^\dagger$ is the auxiliary spinon fermion carrying spin σ (the physical quasiparticle in the Kondo limit), and $b^\dagger$ is the slave boson enforcing no double occupancy. The parameter Q fixes the constraint normalization: $Q = 1$ gives the physical spin-1/2 infinite-U constraint, while $Q = N q_{\rm occ}$ denotes the large-N or channel-rescaled normalization. All saddle-point amplitudes quoted below are interpreted in the same normalization. The constraint is imposed via a Lagrange multiplier δ_c. At the saddle point $b \to b_c \in \mathbb{C}$, the phase of b_c is an internal $U(1)$ gauge choice. We therefore work in the gauge-fixed convention in which observables depend on the amplitude $r \equiv |b_c|$, and the spinon hybridization scale is rW_k.

The renormalized impurity level and gauge-invariant SBMF scale are then

$$\tilde{\epsilon}_\xi = \epsilon_\xi + \delta_c + \delta_\Sigma, \tag{60}$$

$$\tilde{\beta} = |\beta_0|\,|b_c| = |\beta_0|\, r \tag{61}$$

Here δ_Σ is the real principal-value one-body shift of the projected impurity level; if this shift is absorbed into the bare level, one may set $\delta_\Sigma = 0$ and reinterpret ϵ_ξ accordingly. The projected channels entering $H_{\rm ss}$ are defined by Eq. (48); in particular, the hybridization is $V_{\rm eff} = rW$

rather than $c_V\tilde{\beta}$. The equations below define the large-N saddle. For the fixed-W small-width path of Eq. (50), a seed-independent finite-r solution is found at $T_{\rm b} = 10^{-6}$ throughout $0.35 \le \beta_0 \le 0.55$: r decreases smoothly from 0.0106078 to 0.00949995, with maximum constraint and stationarity residuals 6.54×10^{-8} and 4.86×10^{-10}, respectively. The maximum grid-refinement error is 1.71×10^{-6}, and charge continuity is satisfied to 3.66×10^{-23}. At the manuscript reference temperature $T_{\rm b} = 0.1$, however, the same $U \to \infty$ closure has no finite-r branch and terminates at the $r = 0$ boundary. The spectra displayed below therefore use stated frozen values and are not used to claim a finite-temperature enhancement of r. The complete convergence and seed-independence audit is given in the Supplemental Material.

B. Saddle-Point Equations and Gauge-Invariant Amplitude

The Keldysh saddle-point conditions follow by varying the complete contour action with respect to $r = |b_c|$ and δ_c (Supplemental Material):

$$0 = r^2 + n_f - Q, \qquad n_f = -i\int \frac{d\omega}{2\pi}\mathrm{Tr}_s G^<_{ff}, \tag{62}$$

$$0 = 2r\delta_c + \mathcal{F}_W + 2rD_0\,\mathcal{X} + 2r\mathcal{I}_\Gamma. \tag{63}$$

The hybridization force $\mathcal{F}_W$ is linear in rW; $\mathcal{X}$ is the coherent-channel force associated with the unrenormalized projected coefficient $D_0(\beta_0)$; and $\mathcal{I}_\Gamma$ is the full retarded/advanced/ lesser influence-kernel force. Its Langreth form contains both $K^R_\Gamma G^<$ and $K^<_\Gamma G^A$; dropping the latter is not a valid thermal saddle. In the chiral basis the relative dissipative kernel is off-diagonal, so $G^<_{+-}$ and $G^<_{-+}$ cannot be discarded. The Supplemental Material gives the explicit expressions and the no-double-counting rule.

A central point, emphasized to avoid any gauge ambiguity, is that the physical amplitude $r = |b_c|$ is gauge-invariant, whereas the phase of b_c is an internal U(1) slave-boson gauge degree of freedom ($b \to be^{i\theta}$, $f_\sigma \to f_\sigma e^{-i\theta}$). Consequently no observable conclusion is drawn from the phase of b_c. All spectra, condition numbers, and scale estimates below use the gauge-invariant amplitude $\tilde{\beta} = |\beta_0|r$ for the symmetric finite-drive diagnostics. If a numerical saddle is represented with a complex b_c, that phase is only a convention for the auxiliary fields and is not used as a physical source of the resonance shift.

C. Lesser Green's Function and Thermal Occupation

The physical lesser Green's function is fixed by the thermal bath. For a bilinear impurity problem coupled to a reservoir at temperature $T_{\rm b}$, the Keldysh construction gives

$$\begin{aligned} G^<_{\rm bath}(\omega) &= G^R(\omega)\,\Sigma^<_{\rm bath}(\omega)\,G^A(\omega) \\ &= f(\omega)\,[G^A(\omega) - G^R(\omega)]. \end{aligned} \tag{64}$$

with $f(\omega) = [e^{\omega/T_{\rm b}} + 1]^{-1}$. This bath-controlled object is the $G^<$ used for the physical occupations, with $n_f = -i\,\mathrm{Tr}\,G^<(t,t)$. Equivalently, $\Sigma^<_{\rm bath} = if\Gamma_{\rm bath}$ and $i(G^R - G^A) = G^R\Gamma_{\rm bath}G^A$. This sandwich form retains the complete channel matrix and introduces no separate biorthogonal normalization factor.

D. Shifted Friedel-Type Resonance Condition

In the standard Hermitian slave-boson mean field at particle-hole symmetry ($\epsilon_\xi = -U/2$), the Kondo resonance is pinned to the Fermi level, $\omega_{\rm res} = 0$, because the two spin occupations are equal and the saddle point fixes $\tilde{\epsilon}_\xi = 0$. In the present driven problem the coarse-grained self-energy produces spin-selective imaginary shifts $x_\sigma = \zeta_\sigma\Gamma_{\rm PT}$ and therefore different effective widths in the two spin channels. The physical resonance position is then most conservatively described by the gauge-invariant condition

$$\omega_{\rm res} \simeq \mathrm{Re}(E_{\rm EP}) \simeq \mathrm{Re}(\tilde{\epsilon}_\xi), \tag{65}$$

valid when the low-energy line shape remains approximately Lorentzian. This is a shifted resonance condition rather than a claim that the gauge-dependent phase of b_c is observable. The shift is controlled by the renormalized level $\tilde{\epsilon}_\xi$, the spin-dependent widths, and the self-consistent amplitude $\tilde{\beta} = |\beta_0||b_c|$. In the strongly non-Lorentzian regime the observed peak need not coincide exactly with this reference, and we therefore treat the deviation as a line-shape effect rather than as a violation of a sum rule. In the frozen kernel used for the plotted spectra, the local-moment reference has $\mathrm{Re}(\tilde{\epsilon}_\xi) \simeq -1.0$ for $\epsilon_\xi = -U/2 = -1.0$; a fully self-consistent finite-temperature SBMF closure, when a finite-r branch exists, would replace this by its computed saddle-point value.

E. Near-EP Analytic Screening-Scale Estimate

Near an isolated EP, the individual spectral projectors grow as the left and right eigenvectors coalesce, while the complete resolvent remains the finite Jordan expression derived in Sec. IX. We use the condition number $\kappa_{\rm imp}$ only as a diagnostic of nonnormality and EP distance. The screening estimate is controlled instead by the eigenvalues of the complete channel-projected Schrieffer–Wolff matrix. We write

$$T_K^{\rm est} = D_{\rm uv}\exp\left[-\frac{1}{\chi_K j_{\rm scr}}\right], \tag{66}$$

$$j_{\rm scr} = \max\left\{\mathrm{Re}\,j_\alpha :\ j_\alpha \in \mathrm{spec}(\rho_b J^{\rm bio}),\ \mathrm{Re}\,j_\alpha > 0\right\}. \tag{67}$$

Here j_α are the eigenvalues of the dimensionless biorthogonal exchange matrix $\rho_b J^{\rm bio}$, and only positive-real-part eigenvalues are retained as antiferromagnetic screening channels. The coefficient $\chi_K > 0$ fixes the conventional scaling normalization; all plotted estimates use $\chi_K = 1$. A scale increase is possible only if the complete EP-active channel calculation increases such a positive screening eigenvalue. No individual projector residue is inserted. By contrast, a hybridization induced EP generated by a negative scalar product $V^R V^L < 0$ can produce strong spectral nonnormality and a frozen non-Hermitian crossover scale, but it should not automatically be identified with an enhanced thermodynamic Kondo temperature. Supplemental Material gives the corresponding frozen-model scale estimate and explains why the spin-selective sign $x_\uparrow = -x_\downarrow$ alone does not split the scale unless the channel widths, real detunings, or exchange eigenvalues are unequal.

VI. SPECTRAL PROPERTIES AND PHYSICAL REGIMES

Numerical conventions are stated separately for each calculation. The normalized Fig. 2 tests only the constant $k = 0$ kernel. The saddle uses $T_{\rm b} = 10^{-6}$ (with a negative control at $T_{\rm b} = 0.1$), whereas the causal scattering/RG gate uses $D = 0.10$, $G = 0.20\beta_0$, and $\bar{\Gamma} = 0.12$. No cutoff, temperature, or interaction value from one calculation is silently transferred to another.

Figure 2 displays the dynamical fingerprints of the exact compensated core and its relative-gain–loss-only diagnostic limit. The representative coefficients place a symmetric local EP pair at $|\beta_0| = 0.5$, accompanied by a condition-number peak and by a crossover from oscillatory to overdamped local dynamics. Removing the coherent splitting channel ($\Delta_{\rm eff} = 0$, Fig. 2(d)–(f)) removes the coherent local impurity EP mechanism. The remaining passive relative gain–loss dynamics is retained only as a diagnostic contrast: it should not be interpreted as a separate positive-screening Kondo scale enhancement.

a. Condition-number convention. All condition numbers shown below are computed from the Euclidean right-eigenvector matrix of the full 4×4 channel-projected kernel at the plotted k slice, using a single normalization convention for all values of β_0. We use $\kappa_{\rm imp}$ to locate finite-$|\beta_0|$ eigenvector coalescence and to track the distance to the EP. Its absolute magnitude is not used as a standalone physical observable or as an independent definition of the Kondo scale; the physical diagnostics are the spectral line shapes, low-energy scale estimates, occupied fluctuation–dissipation-relation (FDR)-weighted spectra, and frozen channel-projected amplitudes extracted with the same gauge-invariant convention $\tilde{\beta} = |\beta_0||b_c|$.

A. Causal Spectral Concentration and Weak-Flow Control

The complete causal channel density and the channel-resolved weak-coupling flow provide supporting tests of the analytic construction; the full panels and adverse controls are shown in Supplemental Figs. S4 and S5. For $\beta_0 = 0.50$, the active/control ratios are 1.0567 at $g_* = 0.30$, 1.0179 at $g_* = 0.40$, 0.9893 at $g_* = 0.50$, and 0.9166 at $g_* = 1$. Thus the larger leading channel-density eigenvalue and early RG trajectory are bounded early-flow effects in the stated window, but they do not support the stronger statement of an enhanced asymptotic Kondo temperature. The matrix density is positive in the retained passive window, and the result is stable under RG-step refinement. A rate-matched off-EP control can exceed the EP in the largest proper-delay peak, so the Smith delay is not counted as independent evidence of enhancement; this control is shown in the Supplemental Material.

VII. FLUCTUATION-DISSIPATION RELATION AND SPECTRAL FUNCTIONS

For the complete reservoir-coupled kernel, the physical lesser Green's function is fixed by the bath self-energy. Supplemental Fig. S6 verifies that the impurity and bath fluctuation–dissipation relation (FDR) ratios follow the same thermal Fermi function on spectral support. No mode-by-mode occupation of the non-Hermitian eigenvectors is used in the physical FDR, in the occupations, or in the saddle-point equations.

a. Bath-based construction (physical). The physical lesser Green's function is fixed by the thermal bath self-energy. Here this is an instantaneous-equilibrium closure at leading adiabatic order. Corrections are of order $O[(\Omega/\epsilon_O)^2]$ (and of the corresponding gradient order) and would describe drive-induced energy redistribution; no pumping statement is made at the retained order. For a bilinear impurity–bath coupling,

$$G^<_{\rm phys}(\omega) = G^R(\omega)\,\Sigma^<_{\rm bath}(\omega)\,G^A(\omega). \tag{68}$$

If the bath is thermal, then

$$\begin{aligned}\Sigma^<_{\rm bath}(\omega) &= i f(\omega)\Gamma_{\rm bath}(\omega)\\ &= f(\omega)\left[\Sigma^A_{\rm bath}(\omega) - \Sigma^R_{\rm bath}(\omega)\right].\end{aligned} \tag{69}$$

Combining this with the Dyson identity for the bath-induced broadening gives

$$G^<_{\rm phys}(\omega) = f(\omega)\,[G^A(\omega) - G^R(\omega)]. \tag{70}$$

Thus the physical impurity FDR is inherited from the reservoir and is the $G^<$ used in the SBMF self-consistency. Charge conservation is checked through the steady-state collision integral

$$\mathcal{I}_N = \int \frac{d\omega}{2\pi}\,{\rm Tr}\big[\Sigma^< G^> - \Sigma^> G^<\big] = 0, \tag{71}$$

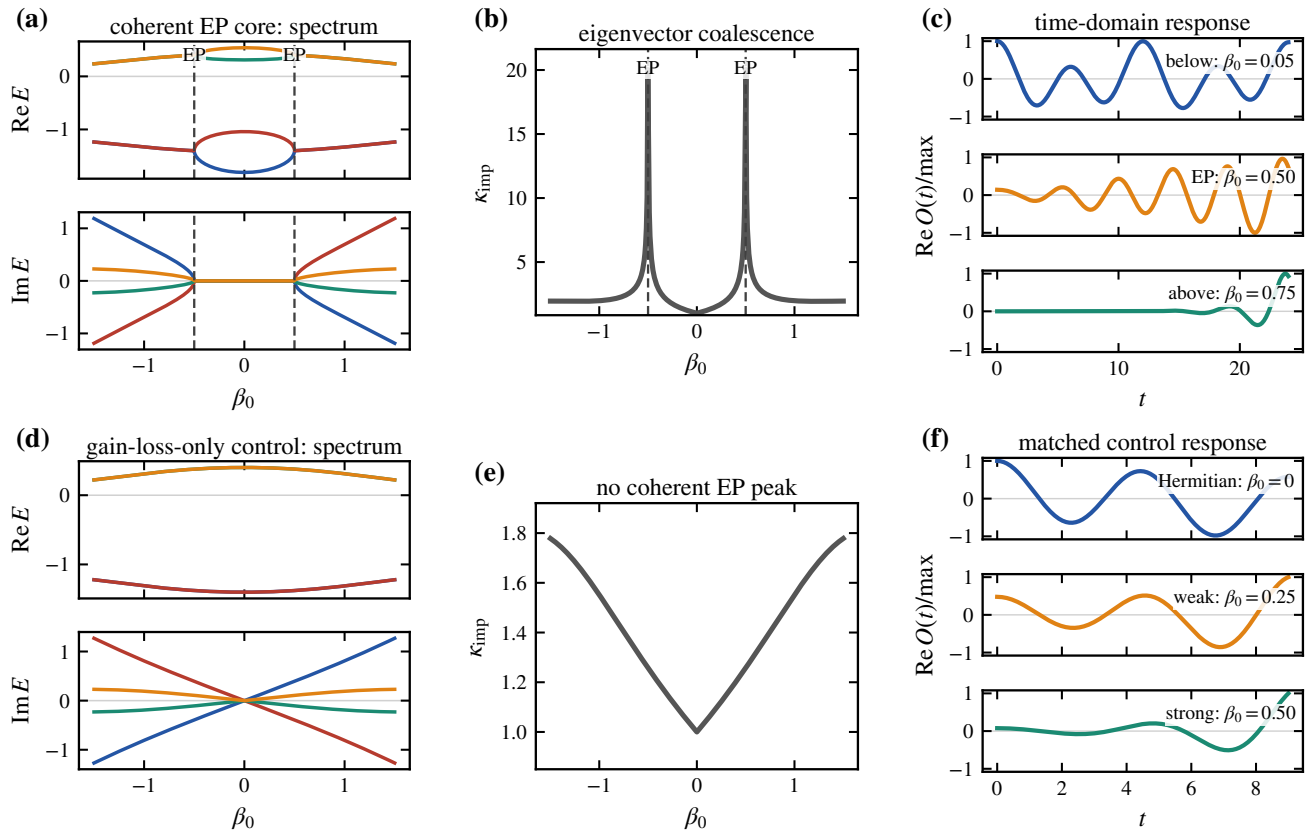


FIG. 2. **Frozen 4×4 compensated-core EP and matched control.** All panels use $r = 1$, $\tilde{\epsilon}_\xi = -1$, fixed scalar hybridization $W = 0.75$, and $G = \Gamma_{\rm PT} = |\beta_0|$ at $k = 0$; hence no physical DOS, cutoff, temperature, or interaction parameter enters. Panels (a)–(c) use the coherent splitting $D = \Delta_{\rm eff} = 0.50$. (a) Real and imaginary parts of all four eigenvalues. The exact local-core EPs are at $\beta_0 = \pm 0.50$. (b) The Euclidean right-eigenvector condition number. The plotted spacing $\Delta\beta_0 = 0.0075$ places its nearest positive sample at 0.5025, where $\kappa_{\rm imp} = 20.0304$; the exact origin remains $\kappa_{\rm imp}(0) = 1$. (c) Separately stacked survival-amplitude traces below, at, and above the EP avoid curve and label overlap. Panels (d)–(f) repeat the same tests after switching off the coherent splitting ($D = 0$). The smooth condition number in (e) and the matched traces in (f) form the relative-gain–loss-only control; they are not interpreted as a coherent impurity EP or as a Kondo-scale enhancement.

which vanishes for the same complete thermal bath self-energy. This is a nontrivial number-continuity check.

b. Impurity spectral function. For the complete physical kernel, $G^A = (G^R)^\dagger$ and the spectral-function matrix is

$$\begin{aligned}\mathcal{A}(\omega) &= i[G^R(\omega) - G^A(\omega)] \\ &= G^R(\omega)\,\Gamma_{\rm tot}(\omega)\,G^A(\omega), \qquad \Gamma_{\rm tot} \succeq 0.\end{aligned} \tag{72}$$

With $P_{\rm imp}$ the impurity projector, the physical impurity density of states is therefore

$$\rho_{\rm imp}(\omega) = \frac{1}{2\pi}\mathrm{Tr}[P_{\rm imp}\mathcal{A}(\omega)] = -\frac{1}{\pi}\mathrm{Im}\,\mathrm{Tr}[P_{\rm imp}G^R(\omega)]. \tag{73}$$

We verify the equality of these two evaluations and positivity numerically. A signed response of the compensated relative kernel, obtained after common damping is subtracted, is not interpreted as a physical density of states.

c. Gauge convention for spectral functions. The physical self-energy from the coarse-graining is first decomposed into a common bath shift/damping, the passive $\mathcal{PT}$ core of Eq. (35), and the Kramers–Kronig detuning of Eq. (34). The off-shell auxiliary sector supplies the real spin-odd shift, while the retarded SOC matrix bath converts this shift into the relative spin-dependent width $\Gamma_{\rm PT}$ through Eq. (25). The effective Hamiltonian is therefore written in terms of the real, gauge-invariant amplitude $r = |b_c|$ and the projected channels $\Gamma_{\rm PT} = r^2\Gamma^{(0)}_{\rm PT}(\beta_0)$, $\Delta_{\rm eff} = r^2\Delta_0^{(0)}$, and $V_{\rm eff} = rW$. The phase of the auxiliary slave-boson field is not used separately in spectral calculations. With this convention, the same $H_{\rm ss}$ is used for eigenvalues, condition numbers, and spectral functions, and no observable is attributed to the gauge-dependent phase of b_c.

VIII. REAL-TIME DYNAMICS

The real-time diagnostic is the complex survival amplitude

$$O(t) = \langle d_\uparrow | e^{-iHt} | d_\uparrow \rangle, \tag{74}$$

evaluated for the same gauge-fixed frozen kernel used in the spectral plots. Main-text Fig. 2(c),(f) shows $\mathrm{Re}\, O(t)$ as separated small multiples; the associated impurity Bloch trajectories and norm check are retained in Supplemental Fig. S1. This is a non-unitary propagator diagnostic of the local EP; it is not used as a normalized thermal expectation value. For Fig. 2(c) we use $\tilde{\epsilon}_\xi = -1$, $D = 0.50$, fixed $W = 0.75$, and $\beta_{0,\mathrm{core}}^{\mathrm{EP}} = 0.50$. The evolution shows a direct dynamical signature of the EP:

- *Below EP* ($\beta_0 < \beta_{0,\mathrm{core}}^{\mathrm{EP}}$, local splitting real): eigenvalues are non-degenerate; the evolution shows underdamped oscillations with a frequency set by $\mathrm{Re}(E_+ - E_-) = 2\mathrm{Re}\, s$ and a decay rate set by $\mathrm{Im}(E_\pm)$.
- *Above EP* ($\beta_0 > \beta_{0,\mathrm{core}}^{\mathrm{EP}}$, local splitting imaginary): eigenvalues have opposite imaginary parts; the gain mode dominates and the evolution shows overdamped decay.
- *Without coherent splitting* ($\Delta_{\mathrm{eff}} = 0$, Fig. 2(d)–(f)): the isolated relative gain–loss block has eigenvalues $\tilde{\epsilon}_\xi \pm i\Gamma_{\mathrm{PT}}$ and is therefore in the broken sector for $\Gamma_{\mathrm{PT}} \neq 0$. The plotted bounded trajectories are normalized or co-decaying diagnostics of the full bath-coupled kernel, not evidence for conservation of the Euclidean norm.

The contrast between the matched no-flip diagnostic and the separated time-domain traces with coherent splitting demonstrates that both projected impurity channels of $\tilde{H}$—the coherent splitting Δ_{eff} and the relative decay channel $i\Gamma_{\mathrm{PT}}$—are necessary for the full finite-drive EP structure. In the channel-projected parametrization, these channels are measured relative to the common scale $\tilde{\beta}$ but should be viewed as distinct projected self-energy components. Equality of their coefficients is a symmetric representative limit, not a symmetry requirement.

IX. CAUSAL SCATTERING AND LINEARIZED CONTACT ALGEBRA

A. Complete scattering and Jordan response

We use the standard Fisher–Lee and Wigner–Smith constructions only as causal scattering diagnostics of the projected kernel [39–41]; they are not introduced here as new formalisms. The basis labels must be kept explicit. In the original spin basis $\Psi_s = (d_\uparrow, d_\downarrow)^T$, the compensated Markov kernel and its positive rate matrix are

$$\begin{aligned} h_s &= \Delta_{\mathrm{eff}}\sigma_x + i\Gamma_{\mathrm{PT}}\sigma_z - i\bar{\Gamma} I, \\ \Gamma_s &= i(\Sigma_s^R - \Sigma_s^A) = 2(\bar{\Gamma} I - \Gamma_{\mathrm{PT}}\sigma_z) \succeq 0. \end{aligned} \tag{75}$$

The Hadamard matrix U_H that defines the fixed chiral fields $\Psi_\chi = U_H^\dagger \Psi_s$ gives

$$\begin{aligned} h_\chi &= U_H^\dagger h_s U_H \\ &= \Delta_{\mathrm{eff}}\sigma_z + i\Gamma_{\mathrm{PT}}\sigma_x - i\bar{\Gamma} I, \\ \Gamma_\chi &= U_H^\dagger \Gamma_s U_H \\ &= 2(\bar{\Gamma} I - \Gamma_{\mathrm{PT}}\sigma_x). \end{aligned} \tag{76}$$

For either representation $\nu = s, \chi$,

$$S_\nu(\omega) = I - i\Gamma_\nu^{1/2} G_\nu^R(\omega) \Gamma_\nu^{1/2}, \qquad S_\chi = U_H^\dagger S_s U_H, \tag{77}$$

and, in the passive window, S_ν is unitary. The Wigner–Smith matrix

$$Q_\nu(\omega) = -iS_\nu^\dagger(\omega)\partial_\omega S_\nu(\omega) \tag{78}$$

is Hermitian and transforms covariantly. Thus its trace and eigenvalues, like the eigenphases of S, are basis invariant. Individual entries such as S_{+-} are representation dependent and are not used as evidence for enhancement. The basis-resolved Green functions and coefficients are worked out in the Supplemental Material.

For the compensated unbroken Markov core, $s = \sqrt{\Delta_{\mathrm{eff}}^2 - \Gamma_{\mathrm{PT}}^2} \in \mathbb{R}$ and the two causal scattering phases give

$$\begin{aligned} \mathrm{Tr}\, Q(\tilde{\epsilon}_\xi) &= \frac{4\bar{\Gamma}}{s^2 + \bar{\Gamma}^2}, \\ \frac{\mathrm{Tr}\, Q_{\Gamma_{\mathrm{PT}}}(\tilde{\epsilon}_\xi)}{\mathrm{Tr}\, Q_0(\tilde{\epsilon}_\xi)} &= \frac{\Delta_{\mathrm{eff}}^2 + \bar{\Gamma}^2}{\Delta_{\mathrm{eff}}^2 - \Gamma_{\mathrm{PT}}^2 + \bar{\Gamma}^2} > 1. \end{aligned} \tag{79}$$

Thus, at fixed Δ_{eff} and $\bar{\Gamma}$, the invariant scattering phase density is analytically concentrated at the resonance center as the EP is approached. Its integrated weight remains fixed, and linewidth- or rate-matched adverse controls need not share the peak ordering. We call Eq. (79) path-specific spectral concentration, not an enhanced thermodynamic Kondo temperature or a basis-independent amplification factor.

The cancellation responsible for its regularity is an exact matrix identity. For the traceless core $h = \Delta_{\mathrm{eff}}\sigma_x + i\Gamma_{\mathrm{PT}}\sigma_z$, $h^2 = s^2 I$, write $a = z - \epsilon_0 - \Sigma_0^R(z)$. Although the individual biorthogonal projectors $P_\pm = (I \pm h/s)/2$ grow as $1/s$, their complete resolvent is

$$G^R(z) = \frac{P_+}{a - s} + \frac{P_-}{a + s} = \frac{aI + h}{a^2 - s^2}. \tag{80}$$

At the EP, $h_{\mathrm{EP}}^2 = 0$, and

$$(aI - h_{\mathrm{EP}})^{-1} = \frac{I}{a} + \frac{h_{\mathrm{EP}}}{a^2}, \tag{81}$$

so the coalescence produces a finite-coefficient simple-plus-double-pole Jordan response, with no additional $1/s$ normalization multiplier. This cancellation is proved by Eq. (80); it is not inferred from NRG. An independent weak-coupling finite-U biorthogonal NRG audit resolves the same matrix direction in branch-tracked transition residues: the channel trace is strongly suppressed while the channel-odd and symmetric off-diagonal components retain the nilpotent Jordan structure. This is a matrix-residue and physical-sum-rule consistency test, not a second proof of the algebraic cancellation or a universal interaction-induced pole-splitting law. For the compensated Markovian kernel,

$$\begin{aligned} \operatorname{Tr} Q_{\mathrm{EP}}(\omega) &= \frac{4\bar{\Gamma}}{(\omega - \operatorname{Re} E_{\mathrm{EP}})^2 + \bar{\Gamma}^2}, \\ \int \frac{d\omega}{2\pi} \operatorname{Tr} Q_{\mathrm{EP}} &= 2. \end{aligned} \tag{82}$$

The numerical implementation satisfies scattering unitarity to 1.33×10^{-15} and the trace-delay/determinant identity to 4.97×10^{-14}. A rate-matched off-EP control can nevertheless have a larger maximum proper delay, so this observable diagnoses causal pole coalescence but is not, by itself, evidence of many-body enhancement.

The same cancellation occurs in the finite-U virtual-charge sum. At particle–hole symmetry, with $A = U/2$ and a channel-independent bare vertex W,

$$\begin{aligned} J_{\mathrm{SW}}(s) &= 2W^2 \left[(AI - h)^{-1} + (AI + h)^{-1}\right] \\ &= \frac{4AW^2}{A^2 - s^2} I, \qquad J_{\mathrm{SW}}(0) = \frac{8W^2}{U}. \end{aligned} \tag{83}$$

Thus neither a Green function nor an exchange vertex acquires an independent Petermann or condition-number multiplier. This causal scattering statement does not assume integrability. We first record its exact one-particle phase condition and then isolate the additional low-energy assumptions under which the finite-U contact problem has a genuine multiparticle Yang–Baxter algebra.

B. Low-energy contact-algebra and screening scope

The eigenvalues $\mathsf{s}_a = e^{2i\delta_a}$ of the complete unitary scattering matrix give the exact one-particle finite-size condition and phase density

$$\det\left[I - e^{ik\mathcal{L}} S(k)\right] = 0, \qquad \frac{1}{2\pi i} \partial_\omega \ln \det S = \frac{1}{2\pi} \operatorname{Tr} Q. \tag{84}$$

This is a basis-independent one-particle statement, not by itself a multiparticle Bethe ansatz. The Supplemental Material gives the full contact matching. After equal-velocity branchwise linearization, removal of curvature, freezing of the component matrix, and restriction to scalar momentum-independent hybridization, the finite-U two-electron contact matrix is the rational Anderson R matrix in a dressed rapidity. It obeys the ordinary Yang–Baxter equation and is $GL(2,\mathbb{C})$ covariant; the EP then enters as an invertible unipotent boundary twist. The bounded-ring many-body theorem is proved separately in Ref. [35]. Curvature, unequal velocities, causal frequency dependence, and non-scalar projected vertices lie outside that theorem, so no full-momentum integrability is claimed here.

For comparison with the causal weak-flow calculation, the only screening estimate retained is the conventional frozen Schrieffer–Wolff form

$$\begin{aligned} J_a^{\mathrm{SW}} &= 2|W_a|^2 \left(|\tilde{\epsilon}_{d,a}|^{-1} + |\tilde{\epsilon}_{d,a} + U|^{-1}\right), \\ T_{K,a}^{\mathrm{SW(est)}} &\simeq D_{\mathrm{uv},a} \exp\left[-1/(\chi_K \rho_{b,a} J_a^{\mathrm{SW}})\right]. \end{aligned} \tag{85}$$

SOC can change this conditional estimate through a positive channel density, velocity Jacobian, overlap, or ordinary charge denominator. It does not supply an independent $1/s$ factor. Negative or complex exchange eigenvalues are not identified with Kondo screening.

X. SCOPE OF THE NEAR-EP SCALE ESTIMATE

Near the EP, the condition number $\kappa_{\mathrm{imp}} \sim C/|s|$ measures eigenvector nonorthogonality. It is useful for locating the projected-core degeneracy but is not a spectral residue, exchange vertex, or quasiparticle weight. The singular parts cancel in the complete Green function [Eq. (80)] and in the finite-U charge sum [Eq. (83)]. Consequently no Z^{bio} or Petermann multiplier is inserted into a propagator, diagrammatic vertex, or Kondo exponent.

On the exact linearized submanifold, the local U preserves the complexified pseudospin symmetry and therefore does not cut off this algebraic Jordan limit. This does not require a divergent observable: the complete resolvent has finite Jordan coefficients and the charge sum has no standalone $1/s$ multiplier. In the complete driven kernel, curvature, velocity mismatch, frequency dependence, or component-dependent vertices can move or split the EP, but their effect must be calculated microscopically rather than represented by a universal interaction floor.

Equation (85) retains the conventional frozen Schrieffer–Wolff scale estimate and permits a conditional increase relative to a matched Hermitian reference only when the projected Dirac/SOC channel increases a positive screening width or exchange eigenvalue. That conditional estimate is conceptually distinct both from multiplying a Hermitian scale by a biorthogonal norm and from the Yang–Baxter identity.

The direct numerical evidence is narrower. The leading eigenvalue of the complete causal channel density is larger in the chosen Fermi window, and the largest exchange eigenvalue initially grows faster than in the matched Hermitian control. Supplemental Fig. S5 shows that

this ordering reverses before strong coupling. The low-temperature slave-boson saddle in Supplemental Fig. S2 validates the r^2 scaling and charge constraint, but the same closure has no finite-r branch at $T_{\rm b} = 0.1$. We therefore describe only bounded spectral concentration and an early-flow ordering on the displayed path. The estimate in Eq. (85) remains conditional on a positive screening channel, and no enhanced thermodynamic Kondo temperature is claimed.

XI. CONCLUSIONS

We have shown that a strictly Hermitian, periodically driven Dirac impurity admits an emergent passive non-Hermitian retarded kernel through two linked steps: off-shell angular harmonics produce a real spin-odd shift, and the SOC matrix bath converts it into relative decay. The causal finite-band problem also clarifies the limits of the constant-core picture. The normalized oscillatory-drive benchmark exhibits avoided coalescence, whereas a distinct stationary rotating-drive construction supports a continuous family of full-momentum causal pole EPs. Their double zeros are protected by local winding and pole exchange. On the representative finite systems, the complete-basis matrix block-Wilson calculations preserve this topology at $U/\bar{\Gamma}_{\rm self} = 1.225$. Independently reoptimized $N = 10$ and 12 chains share the same pole exchange and zero count, with the common-contour Rouché ratio $0.6775 < 1$. This establishes finite-interaction survival and finite-N persistence over the stated pair, not a thermodynamic-limit full-density-matrix NRG fixed point.

At fixed nonzero spectral detuning, the complete response has no standalone eigenvector-normalization divergence as the EP is approached. Divergent individual biorthogonal projectors combine into the finite-coefficient Jordan simple-plus-double-pole resolvent, and the particle–hole-symmetric finite-U charge resolvent has the same cancellation. The physical double pole remains; what is absent is an additional Petermann or condition-number multiplier. The Wigner–Smith trace therefore provides a causal pole and linewidth fingerprint rather than an independent enhancement factor. A separate operator-resolved NRG audit preserves the nilpotent residue direction in its validated early-iteration window, while the weak-flow and saddle calculations establish neither a universal interaction law nor an enhanced thermodynamic Kondo temperature.

Finally, branchwise linearization identifies a precise integrable submanifold. Its finite-U contact matrix is the rational Anderson matrix in a dressed rapidity and is $GL(2,\mathbb{C})$ covariant. Thus the off-EP metric similarity preserves the YBE, while the EP is represented by an invertible unipotent boundary twist. Curvature and the energy-dependent driven self-energy lie outside this theorem. Accordingly, neither full-momentum Yang–Baxter integrability nor inheritance of the Markov-core many-body Jordan amplification by the causal pole EP is asserted.

CODE AND DATA AVAILABILITY

The numerical scripts and tabulated data used to generate the main and Supplemental figures are included in the accompanying reproducibility archive. The archive contains independent checks of the finite-U contact algebra, the rotating-drive two-cut EP and SOC continuation, and the complete-basis finite-chain topology used in Fig. 1. It also contains the equation-of-motion identity, kept-space calibration, branch tracking, no-truncation regression, the $N = 10 \to 12$ common-contour Rouché audit, and scripts needed to reproduce the main and Supplemental figures. A version-controlled copy of the code, data, figure outputs, and manuscript sources is available in the GitHub repository `VMKPHYSMATH/emergent-pt-dirac-impurity` [46]. The current public snapshot is tagged `v2.0.1`; the all-versions archival record is doi:10.5281/zenodo.21434682. The full-momentum certificate introduced in this revision is included in the submission archive and will be incorporated into the next public archival snapshot.

SUPPLEMENTAL MATERIAL

Detailed derivations of the auxiliary-mode influence functional, Keldysh mean-field equations, physical fluctuation–dissipation construction, complete biorthogonal and finite-U charge-resolvent identities, causal matrix-RG validation, the finite-U two-electron contact algebra, and parameter tables are provided in the Supplemental Material [47]. The main text remains self-contained by stating the effective projected kernel, physical spectrum, and claim boundary explicitly.

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[47] (2026), see Supplemental Material at [URL will be inserted by publisher] for detailed Wigner and Schur-complement derivations, Keldysh and screening constructions, low-temperature saddle, causal scattering validation, finite-U contact algebra, the operator-resolved NRG residue audit, the full-momentum nonlinear-pole certificate, and Figs. S1–S9.